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COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

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Quiz #1

February 5, 2020

Closed Notes and Book

Duration: 20 minutes

$$\left. \begin{array}{l} T_{\infty,1} = 40^\circ\text{C} \\ h_1 = 30 \frac{\text{W}}{\text{m}^2\text{K}} \\ T_{\infty,2} = -20^\circ\text{C} \\ h_2 = 65 \frac{\text{W}}{\text{m}^2\text{K}} \end{array} \right\} \begin{array}{l} L = 4 \times 10^{-3} \text{m} \\ k = 1.4 \frac{\text{W}}{\text{mK}} \end{array}$$

The rear window of an automobile is defogged by passing warm air at  $40^\circ\text{C}$  over its inner surface, and the associated convection coefficient is  $30 \text{ W}/(\text{m}^2\cdot\text{K})$ . Under conditions for which the outside ambient air temperature is  $-20^\circ\text{C}$  and the associated convection coefficient is  $65 \text{ W}/(\text{m}^2\cdot\text{K})$ , what are the inner and outer surface temperatures of the window? The window glass is 4 mm thick and the thermal conductivity of glass is  $1.4 \text{ W}/(\text{m}\cdot\text{K})$ .

$$q'' = \frac{q}{A} = \frac{T_{\infty,1} - T_{\infty,2}}{\frac{L}{k} + \frac{1}{h_1} + \frac{1}{h_2}} = \frac{(40 - (-20))^\circ\text{C}}{\frac{4 \times 10^{-3} \text{m}}{1.4 \frac{\text{W}}{\text{mK}}} + \frac{1}{30 \frac{\text{W}}{\text{m}^2\text{K}}} + \frac{1}{65 \frac{\text{W}}{\text{m}^2\text{K}}}}$$

Assume Steady State:

$$= \frac{60^\circ\text{C}}{\frac{4 \times 10^{-3} \text{m}}{1.4 \frac{\text{W}}{\text{mK}}} + \frac{1}{30 \frac{\text{W}}{\text{m}^2\text{K}}} + \frac{1}{65 \frac{\text{W}}{\text{m}^2\text{K}}}}$$

$$q'' = h_1(T_{\infty,1} - T_i) = k \frac{T_i - T_o}{L} = h_2(T_o - T_{\infty,2})$$

$$\Rightarrow T_i = T_{\infty,1} - \frac{q''}{h_1} = 40^\circ\text{C} - \left( \frac{60^\circ\text{C}}{\left( \frac{4 \times 10^{-3} \text{m}}{1.4 \frac{\text{W}}{\text{mK}}} + \frac{1}{30 \frac{\text{W}}{\text{m}^2\text{K}}} + \frac{1}{65 \frac{\text{W}}{\text{m}^2\text{K}}} \right)} \right) \cdot 30 \frac{\text{W}}{\text{m}^2\text{K}}$$

$$\Rightarrow T_o = T_{\infty,2} + \frac{q''}{h_2} = -20^\circ\text{C} + \left( \frac{60^\circ\text{C}}{\left( \frac{4 \times 10^{-3} \text{m}}{1.4 \frac{\text{W}}{\text{mK}}} + \frac{1}{30 \frac{\text{W}}{\text{m}^2\text{K}}} + \frac{1}{65 \frac{\text{W}}{\text{m}^2\text{K}}} \right)} \right) \cdot 65 \frac{\text{W}}{\text{m}^2\text{K}}$$